

1. a) Find all solutions of $e^{2z-1}=i$.
 b) Let $z_1, z_2 \in \mathbb{C}$. Show that $\frac{z_1-z_2}{2\pi i} \in \mathbb{Z}$ is true, if $e^{z_1} = e^{z_2}$.
2. In each part, determine the image of the set under the given mapping and sketch in the plane.

a) Sketch the following region in the complex plane

$$2 < |z| < 4, -\frac{2\pi}{3} < \arg(z) < \frac{2\pi}{3}$$

and

Find the image of the given region under the mapping $\omega = \operatorname{Ln} z$ and sketch it.

b) Sketch the following region in the complex plane

$$1 \leq x \leq 2, \frac{\pi}{6} \leq y \leq \frac{\pi}{2}$$

and

Find the image of the given region under the mapping $\omega = e^z$ and sketch it.

3. Expand $f(z) = \frac{e^z}{(z+1)^4}$ in a series in powers of $z + 1$ valid for $0 < |z + 1| < \infty$.

By using this series, determine the type of singularity at $z = -1$.

Find the Residue of $f(z) = \frac{e^z}{(z+1)^4}$ at $z = -1$ by using the series.

4. Using contour integration, evaluate $\int_{\Gamma} \bar{z} dz$, where Γ is the contour consisting of three straight line segments connecting the points $(1,0), (0,2), (-2,0), (1,0)$ traversed in a counter clockwise direction.

5. a) Evaluate $\int_C \frac{z^3}{(z+1)^3} dz$, where C is the positively oriented circle of radius 2, center $(0,0)$.

- b) Evaluate $\int_C \frac{z+1}{z^2+1} dz$, where $C: 2|z-1|=1$ with negative orientation.

IN EACH PART, WRITE YOUR SUPPORTING REASONS EXPLICITLY!

6. a) Show that $\int_0^{2\pi} \frac{d\theta}{1-2a\cos\theta+a^2} = \frac{2\pi}{1-a^2}$ by using Residue, where $|a| < 1$.

b) Compute the principal value of $\int_{-\infty}^{\infty} \frac{x.e^{2ix}}{x^2-1} dx$.

$$1) a) e^{2z-1} = i \quad z = x + iy$$

$$\Rightarrow e^{2(x+iy)-1} = e^{2x-1} e^{2iy} = e^{i\frac{\pi}{2}}$$

$$\Rightarrow x = \frac{1}{2} \quad y = \frac{\pi}{4} + n\pi \Rightarrow \boxed{z = \frac{1}{2} + \frac{\pi i}{4} + n\pi i \quad n = 0, \pm 1, \pm 2, \dots}$$

$$b) e^{z_1} = e^{z_2} \Rightarrow \frac{e^{z_1}}{e^{z_2}} = 1 \Rightarrow e^{z_1 - z_2} = 1 = e^{2n\pi i}$$

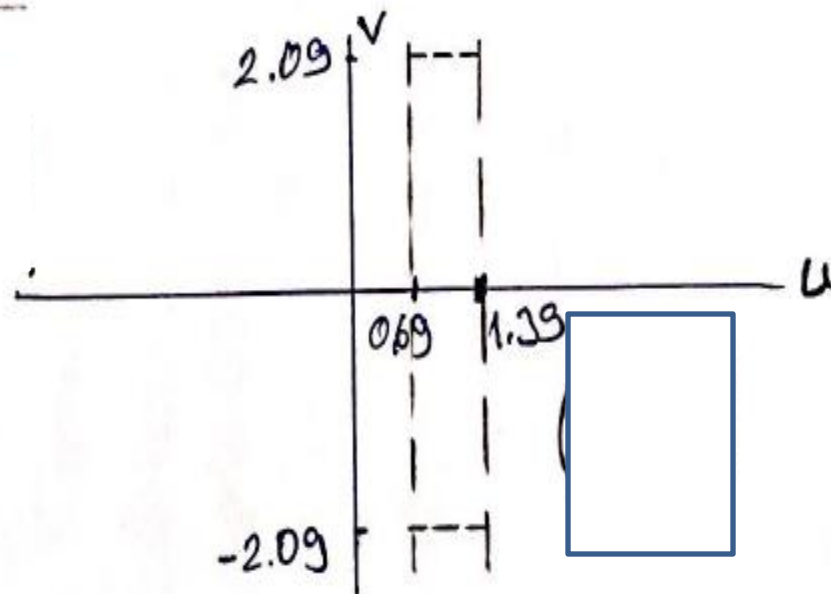
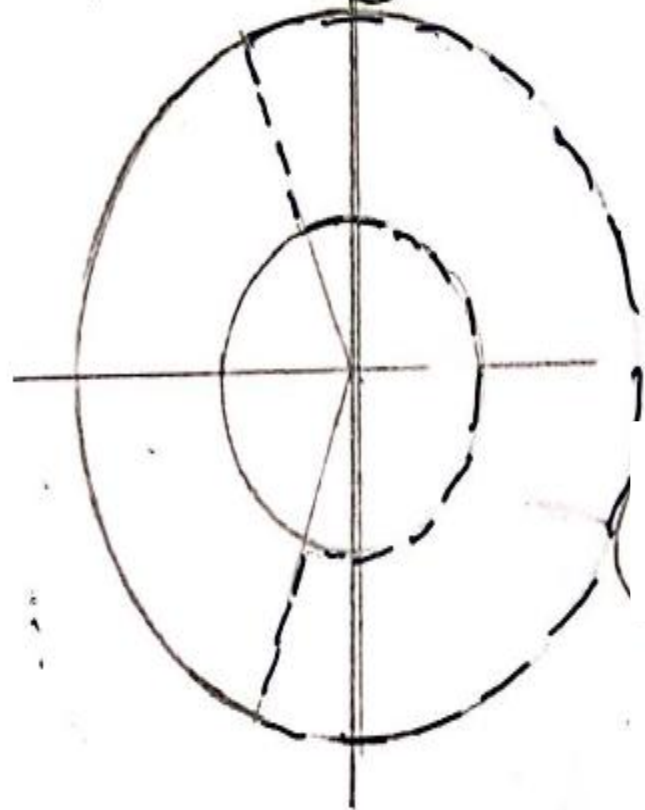
$$\Rightarrow z_1 - z_2 = 2n\pi i \quad n \in \mathbb{Z}$$

$$\frac{z_1 - z_2}{2\pi i} = \frac{2n\pi i}{2\pi i} = n \in \mathbb{Z} \text{ olur.}$$

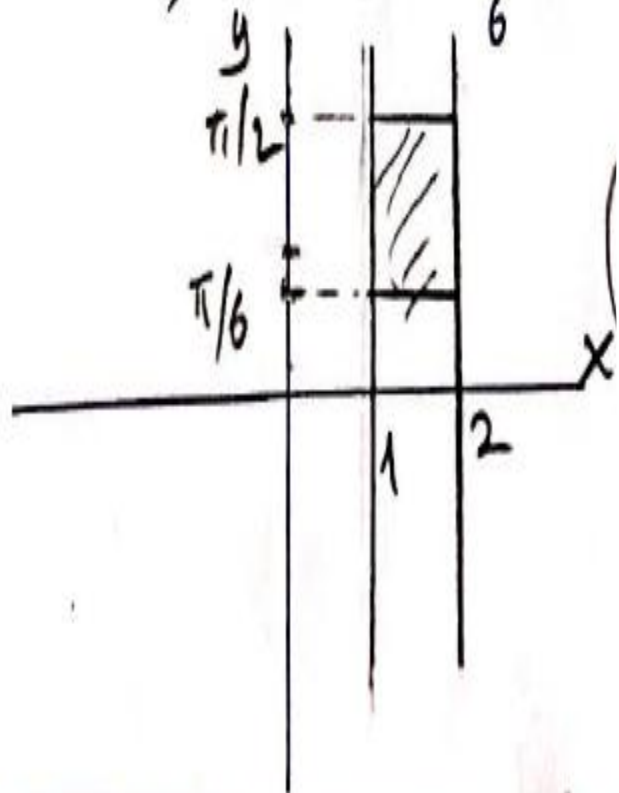
2) a) $2 < |z| < 4$ $-\frac{2\pi}{3} < \arg(z) < \frac{2\pi}{3}$ $\omega = \ln z$

$$\log_e 2 < u < \log_e 4 \Rightarrow 0.69 < u < 1.39$$

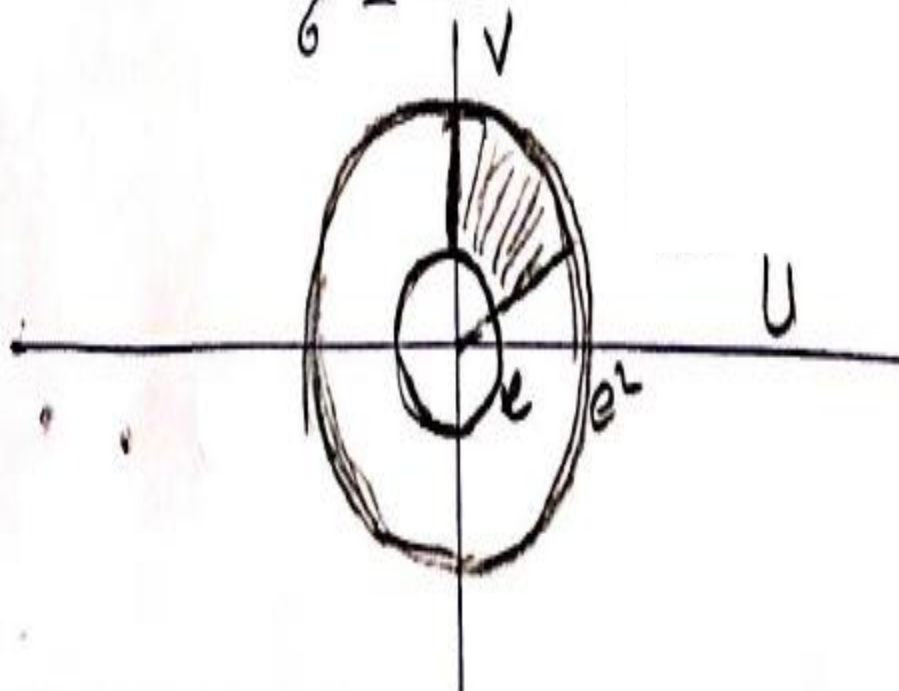
$$-\frac{2\pi}{3} < v < \frac{2\pi}{3} \Rightarrow 2.09 < v < -2.09$$



$$b) 1 \leq x \leq 2 \quad \frac{\pi}{6} \leq y \leq \frac{\pi}{2} \quad w = e^z \Rightarrow e^1 \leq |w| \leq e^2$$



$$\frac{\pi}{6} \leq \arg(w) \leq \frac{\pi}{2}$$



$$3) \quad f(z) = \frac{e^z}{(z+1)^4} \quad 0 < |z+1| < \infty$$

$$z+1=u \Rightarrow z=u-1$$

$$f(u) = \frac{e^{u-1}}{u^4} = \frac{1}{u^4} e^{-1} e^u = \frac{1}{e u^4} \left(1 + u + \frac{u^2}{2!} + \frac{u^3}{3!} + \frac{u^4}{4!} + \frac{u^5}{5!} + \dots \right)$$

$$= \frac{1}{e} \left(\frac{1}{u^4} + \frac{1}{u^3} + \frac{1}{2u^2} + \frac{1}{6u} + \frac{1}{4!} + \frac{u}{5!} + \frac{u^2}{6!} + \dots \right) \quad 0 < u < \infty$$

$$f(z) = \frac{1}{e} \left(\frac{1}{(z+1)^4} + \frac{1}{(z+1)^3} + \frac{1}{2(z+1)^2} + \frac{1}{6(z+1)} + \frac{1}{4!} + \frac{z+1}{5!} + \frac{(z+1)^2}{6!} + \dots \right)$$

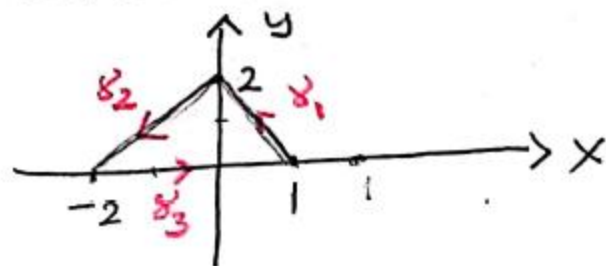
$$0 < |z+1| < \infty$$

Serinin esas kısmı $(z+1)^{-4}$ ile başladığından
 $z=-1$ noktası 4. mertebeden kutuptur.

$$\text{Res}(f(z), -1) = \frac{1}{6e}$$

4.

Using contour integration, evaluate $\int_{\Gamma} \bar{z} dz$ where Γ is the contour consisting of three straight line segments connecting the points $(1,0)$, $(0,2)$, $(-2,0)$ traversed in a counter-clockwise direction.



$$\gamma_1 : 1 \rightarrow 2i$$

$$\gamma_2 : 2i \rightarrow -2$$

$$\gamma_3 : -2 \rightarrow 1$$

$$z_1(t) = (2i-1)t + 1, \quad t : 0 \rightarrow 1$$

$$z_2(t) = (-2-2i)t + 2i, \quad t : 0 \rightarrow 1$$

$$z_3(t) = t, \quad t : -2 \rightarrow 1$$

$$\begin{aligned} \int_{\gamma_1} \bar{z} dz &= \int_0^1 [(-2i-1)t + 1](2i-1) dt = \int_0^1 [(2i-1) + t((-1)^2 - (2i)^2)] dt \\ &= (2i-1)t \Big|_0^1 + 5 \int_0^1 t dt = 2i-1 + 5 \cdot \frac{1}{2} = 2i + \frac{3}{2} \end{aligned}$$

$$\begin{aligned}\int_{\gamma_2} \bar{z} dz &= \int_0^1 ((-2+2i)t - 2i) \cdot (-2+2i) dt \\ &= \int_0^1 8i t dt + \int_0^1 (+4i - 4) dt = \boxed{4i}\end{aligned}$$

$$\int_{\gamma_3} \bar{z} dz = \int_{-2}^1 t dt = \frac{t^2}{2} \Big|_{-2}^1 = \frac{1}{2} - \frac{(-2)^2}{2} = \boxed{-\frac{3}{2}}$$

$$\begin{aligned}\int_{\Gamma} \bar{z} dz &= \int_{\gamma_1} \bar{z} dz + \int_{\gamma_2} \bar{z} dz + \int_{\gamma_3} \bar{z} dz \\ &= 2i + \cancel{\frac{3}{2}} + 4i - \cancel{\frac{3}{2}} = \boxed{6i}\end{aligned}$$

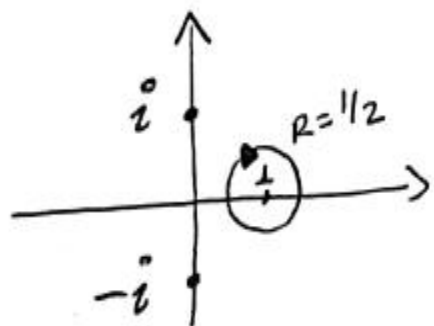
5.

a) Evaluate $\int_C \frac{z^3}{(z+1)^3} dz$, where C is the positively oriented circle of radius 2 about 0.

Since $z = -1$ is inside the circle, the extension of the Cauchy integral formula gives

$$\int_C \frac{z^3}{(z+1)^3} dz = \frac{2\pi i}{2!} \frac{d^2}{dz^2} \cdot z^3 \Big|_{z=-1} = \boxed{6\pi i}$$

b) Evaluate $\int_C \frac{z+1}{z^2+1} dz$, where $C: 2|z-1|=1$ with negative orientation



$$\int_C f(z) dz = \oint_C \frac{z+1}{(z-i)(z+i)} dz = 0$$

f is analytic in C

f is analytic on $\left(C - \{i\} \right)$
 $\left(C - \{-i\} \right)$.

6.

show that
$$\int_0^{2\pi} \frac{d\theta}{1 - 2a \cos \theta + a^2} = \frac{2\pi}{1 - a^2}, \text{ where } |a| < 1$$

Let $C: z = \cos \theta + i \sin \theta \quad 0 \leq \theta \leq 2\pi \Rightarrow$

$$\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right) \quad d\theta = \frac{dz}{iz}$$

$$\int_0^{2\pi} \frac{d\theta}{1 - 2a \cos \theta + a^2}$$

$$= \oint_C \frac{1}{1 - 2a \cdot \frac{1}{2} \left(z + \frac{1}{z} \right) + a^2} \cdot \frac{dz}{iz}$$

$$= \oint_C \frac{i dz}{az^2 - a^2z - z + a}$$

$$= \oint_C \frac{idz}{(z-a)(az-1)} = 2\pi i \cdot i \cdot \operatorname{Res}_{z=a} \frac{1}{(z-a)(az-1)}$$

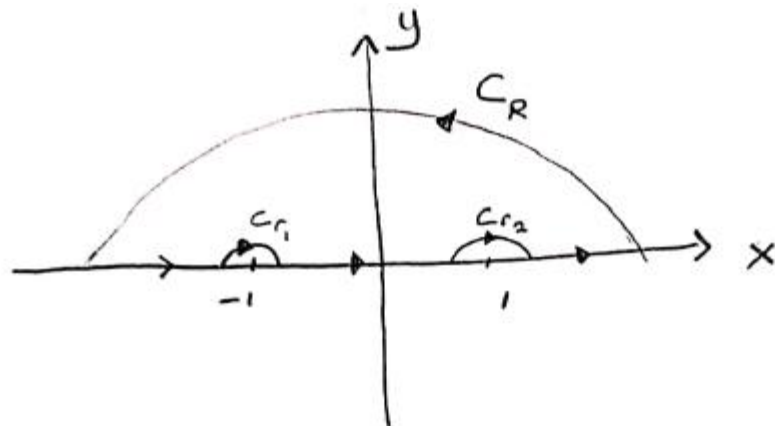
$$= 2\pi \cdot \frac{-1}{a^2-1} = \boxed{\frac{2\pi}{1-a^2}}$$

Compute the principal value of $\int_{-\infty}^{\infty} \frac{x \cdot e^{2ix}}{x^2-1} dx$.

The improper integral has singularities at $x = \pm 1$.

The principal value of the integral is defined to be

$$\lim_{\substack{R \rightarrow \infty \\ r_1, r_2 \rightarrow 0^+}} \left(\int_{-R}^{-1-r_1} + \int_{-1+r_1}^{1-r_2} + \int_{1+r_2}^R \right) \cdot \frac{x \cdot e^{2ix}}{x^2-1} dx.$$



$$\text{Let } I_1 = \int_{C_1} \frac{z e^{2iz}}{z^2 - 1} dz$$

$$I_2 = \int_{C_2} \frac{z e^{2iz}}{z^2 - 1} dz$$

$$I_R = \int_{C_R} \frac{z e^{2iz}}{z^2 - 1} dz$$

$f(z) = \frac{z \cdot e^{2iz}}{z^2 - 1}$ is analytic inside the above closed contour

By the Cauchy integral theorem

$$\left(\int_{-R}^{-1-r_1} + \int_{-1+r_1}^{1-r_2} + \int_{1+r_2}^R \right) \frac{x \cdot e^{2ix}}{x^2 - 1} dx + I_1 + I_2 + I_R = 0$$

since $\frac{z}{z^2-1} \rightarrow 0$ as $z \rightarrow \infty$, so $\lim_{R \rightarrow \infty} I_R = 0$

Since $z = \pm 1$ are simple poles of f ,
 $\lim_{r_1 \rightarrow 0^+} I_1 = -i\pi \cdot \text{Res}(f, -1) = -i\pi \cdot \lim_{z \rightarrow -1} (z+1) \cdot f(z) = (-i\pi) \cdot \frac{e^{-2i}}{-2}$

Similarly, $\lim_{r_2 \rightarrow 0^+} I_2 = -i\pi \cdot \text{Res}(f, 1) = -i\pi \cdot \frac{e^{2i}}{2}$

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{x \cdot e^{2ix}}{x^2-1} dx = i\pi \frac{e^{-2i}}{2} + i\pi \frac{e^{2i}}{2} = \boxed{i\pi \cos 2}$$

$$\textcircled{1} \text{ a) } f(x, y) = \operatorname{Im}(\cos z) \quad \cos z = \frac{e^{iz} + e^{-iz}}{2} \quad z = x + iy$$

$$\Rightarrow \cos z = \frac{1}{2} (e^{i(x+iy)} + e^{-i(x+iy)}) = \frac{1}{2} (e^{ix} e^{-y} + e^{-ix} e^y)$$

$$= \frac{1}{2} [(\cos x + i \sin x) e^{-y} + (\cos x - i \sin x) e^y]$$

$$= \cos x \frac{e^y + e^{-y}}{2} + i \sin x \frac{e^{-y} - e^y}{2} = \cos x \cosh y + i \sin x \sinh y$$

$$f(x, y) = \operatorname{Im}(\cos z) = -\sin x \sinh y$$

$$\begin{aligned} f_x &= -\cos x \sinh y & f_{xx} &= \sin x \sinh y \\ f_y &= -\sin x \cosh y & f_{yy} &= -\sin x \sinh y \end{aligned} \quad \left\{ \begin{array}{l} f_{xx} + f_{yy} = 0 \end{array} \right.$$

$\Rightarrow f(x, y)$ harmonic.

$$b) g(x,y) = \operatorname{Re}(\sinh z) \quad \sinh z = \frac{e^{iz} - e^{-iz}}{2i} \quad z = x + iy$$

$$\sinh z = \frac{1}{2i} (e^{i(x+iy)} - e^{-i(x+iy)}) = -\frac{i}{2} (e^{ix} e^{-y} - e^{-ix} e^y)$$

$$= -\frac{i}{2} [(\cos x + i \sin x) e^{-y} - (\cos x - i \sin x) e^y]$$

$$= \frac{1}{2} [(-i \cos x + \sin x) e^{-y} + (i \cos x + \sin x) e^y]$$

$$= \sin x \frac{e^{-y} + e^y}{2} + i \cos x \frac{e^y - e^{-y}}{2} = \sin x \cosh y + i \cos x \sinh y$$

$$g(x,y) = \operatorname{Re}(\sinh z) = \sin x \cosh y$$

$$g_x = \cos x \cosh y$$

$$g_{xx} = -\sin x \cosh y$$

$$g_y = \sin x \sinh y$$

$$g_{yy} = \sin x \cosh y$$

$$g_{xx} + g_{yy} = 0$$

$\Rightarrow g(x,y)$ harmonik

② a) $C: |z-i|=1$ $z-i=e^{i\theta}$ $z=i+e^{i\theta}$ $dz=ie^{i\theta}d\theta$ $\bar{z}=-i+e^{-i\theta}$

$$\oint_C \bar{z} dz = \int_0^{2\pi} (-i+e^{-i\theta}) ie^{i\theta} d\theta = \int_0^{2\pi} (e^{i\theta} + i) d\theta = \left(\frac{1}{i} e^{i\theta} + i\theta \right)_0^{2\pi} = \frac{1}{i} (e^{2\pi i} - 1) + 2\pi i = 2\pi i$$

b) $z^2+2=0$ $(z+\sqrt{2}i)(z-\sqrt{2}i)=0$ $z_1=-\sqrt{2}i$ $z_2=\sqrt{2}i$

$C: |z-i|=1 \Rightarrow z_2=\sqrt{2}i$ C nin içinde, $z_1=-\sqrt{2}i$ C nin dışında

$$\oint_C \frac{dz}{z^2+2} = \oint_C \frac{\frac{1}{z+\sqrt{2}i}}{z-\sqrt{2}i} dz = 2\pi i f(\sqrt{2}i) = 2\pi i \frac{1}{2\sqrt{2}i} = \frac{\pi}{\sqrt{2}}$$

$$c) f(z) = \frac{1}{z^2+z} = \frac{1}{z(z+1)}$$

$$a) |z| > 1 \quad \left| \frac{1}{z} \right| < 1 \Rightarrow f(z) = \frac{1}{z^2(1+\frac{1}{z})} = \frac{1}{z^2} \sum_{n=0}^{\infty} \left(-\frac{1}{z}\right)^n$$

$$f(z) = \sum_{n=0}^{\infty} (-1)^n \frac{1}{z^{n+2}} \quad |z| > 1$$

$$b) 1 < |z-1| < 2$$

$$f(z) = \frac{1}{z(z+1)} = \frac{A}{z} + \frac{B}{z+1} \Rightarrow A=1 \quad B=-1 \quad f(z) = \frac{1}{z} - \frac{1}{z+1}$$

$$\frac{1}{z} = \frac{1}{z-1+1} = \frac{1}{(z-1)\left(1+\frac{1}{z-1}\right)} = \frac{1}{z-1} \sum_{n=0}^{\infty} \frac{(-1)^n}{(z-1)^n} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(z-1)^{n+1}} \quad |z-1| > 1$$

$$\frac{1}{z+1} = \frac{1}{z-1+2} = \frac{1}{2\left(1+\frac{z-1}{2}\right)} = \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z-1}{2}\right)^n = \sum_{n=0}^{\infty} (-1)^n \frac{(z-1)^n}{2^{n+1}} \quad |z-1| < 2$$

$$\Rightarrow f(z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(z-1)^{n+1}} - \sum_{n=0}^{\infty} (-1)^n \frac{(z-1)^n}{2^{n+1}} \quad 1 < |z-1| < 2$$

④ a) $C: |z|=1 \quad z=e^{i\theta} \quad dz=i e^{i\theta} d\theta \quad d\theta = \frac{dz}{iz} \quad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$

$$\sin\theta = \frac{1}{2i} \left(z - \frac{1}{z} \right) = \frac{1}{2iz} (z^2 - 1)$$

$$\int_0^{2\pi} \frac{d\theta}{3+2\sin\theta} = \oint_C \frac{1}{3 + \frac{1}{iz}(z^2-1)} \frac{dz}{iz} = \oint_C \frac{dz}{z^2 + 3iz - 1}$$

$$z^2 + 3iz - 1 = 0 \Rightarrow z_{1,2} = \frac{-3i \pm \sqrt{-9+4}}{2} = \frac{(-3 \pm \sqrt{5})i}{2}$$

$$z_1 = \frac{-3+\sqrt{5}}{2}i \quad \text{Cinîi içinde,} \quad z_2 = \frac{-3-\sqrt{5}}{2}i \quad \text{Cinîi dışında}$$

$$f(z) = \frac{1}{z^2 + 3iz - 1} \Rightarrow \text{Res } f(z) \Big|_{z=z_1} = \frac{1}{2z+3i} \Big|_{z=z_1} = \frac{1}{(-3+\sqrt{5})i+3i} = \frac{1}{\sqrt{5}i} \checkmark$$

$$\Rightarrow \oint_C \frac{dz}{z^2 + 3iz - 1} = \int_0^{2\pi} \frac{d\theta}{3+2\sin\theta} = 2\pi \cdot \frac{1}{\sqrt{5}i} = \frac{2\pi}{\sqrt{5}} \checkmark$$

b) $\oint_C e^{\frac{2}{1-z}} dz$ $C: |z-1|=1$ $z=1$ C nin içində
təkcə nöqtə

$$e^t = 1 + \frac{t}{1!} + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots$$

$$e^{-\frac{2}{z-1}} = 1 + \frac{-2}{z-1} + \frac{1}{2!} \left(\frac{-2}{z-1} \right)^2 + \dots$$

$$\Rightarrow \operatorname{Res}_{z=1} e^{\frac{2}{1-z}} = -2 \Rightarrow \oint_C e^{\frac{2}{1-z}} dz = (-2) 2\pi i$$

$$= -4\pi i$$

Expand the rational function

$$f(z) = \frac{1 + 2z^2}{z^3 + z^5}$$

around $z = 0$.

Solution: Note that f has a singularity at 0, so we can't expect a convergent Taylor series expansion. We'll aim for the next best thing using the following shortcut.

$$f(z) = \frac{1}{z^3} \frac{2(1 + z^2) - 1}{1 + z^2} = \frac{1}{z^3} \left[2 - \frac{1}{1 + z^2} \right].$$

Using the geometric series we have

$$\frac{1}{1 + z^2} = \frac{1}{1 - (-z^2)} = \sum_{n=0}^{\infty} (-z^2)^n = 1 - z^2 + z^4 - z^6 + \dots$$

Putting it all together

$$f(z) = \frac{1}{z^3} (2 - 1 + z^2 - z^4 + \dots) = \left(\frac{1}{z^3} + \frac{1}{z} \right) - \sum_{n=0}^{\infty} (-1)^n z^{2n+1}$$

Note. The first terms are called the **singular part**, i.e. those with negative powers of z . The summation is called the regular or analytic part. Since the geometric series for $1/(1 + z^2)$ converges for $|z| < 1$, the entire series is valid in $0 < |z| < 1$

Find the Taylor series for

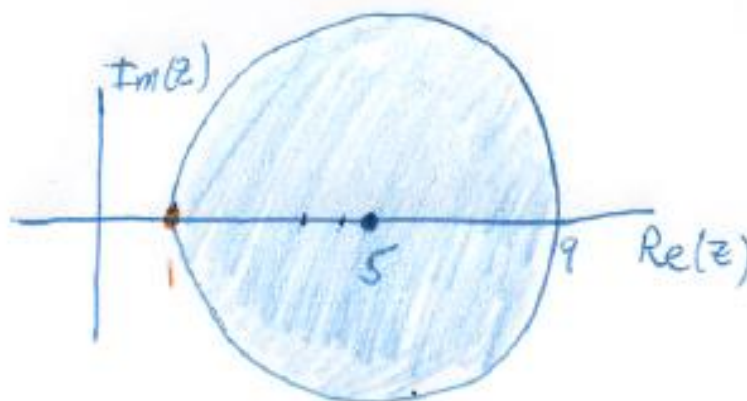
$$f(z) = \frac{1}{1-z}$$

around $z = 5$. Give the radius of convergence.

Solution: We have to manipulate this into standard geometric series form.

$$f(z) = \frac{1}{-4(1 + (z-5)/4)} = -\frac{1}{4} \left(1 - \left(\frac{z-5}{4} \right) + \left(\frac{z-5}{4} \right)^2 - \left(\frac{z-5}{4} \right)^3 + \dots \right)$$

Since $f(z)$ has a singularity at $z = 1$ the radius of convergence is $R = 4$. We can also see this by considering the geometric series. The geometric series ratio is $(z-5)/4$. So the series converges when $|z-5|/4 < 1$, i.e. when $|z-5| < 4$, i.e. $R = 4$.



. Find the Laurent series for

$$f(z) = \frac{z+1}{z}$$

around $z_0 = 0$. Give the region where it is valid.

Solution: The answer is simply

$$f(z) = 1 + \frac{1}{z}.$$

This is a Laurent series, valid on the infinite region $0 < |z| < \infty$.

Find the Laurent series for

$$f(z) = \frac{z}{z^2 + 1}$$

around $z_0 = i$. Give the region where your answer is valid. Identify the singular (principal) part.

Solution: Using partial fractions we have

$$f(z) = \frac{1}{2} \cdot \frac{1}{z-i} + \frac{1}{2} \cdot \frac{1}{z+i}.$$

Since $\frac{1}{z+i}$ is analytic at $z = i$ it has a Taylor series expansion. We find it using geometric series.

$$\frac{1}{z+i} = \frac{1}{2i} \cdot \frac{1}{1 + (z-i)/(2i)} = \frac{1}{2i} \sum_{n=0}^{\infty} \left(-\frac{z-i}{2i} \right)^n$$

So the Laurent series is

$$f(z) = \frac{1}{2} \cdot \frac{1}{z-i} + \frac{1}{4i} \sum_{n=0}^{\infty} \left(-\frac{z-i}{2i} \right)^n$$

The singular (principal) part is given by the first term. The region of convergence is $0 < |z-i| < 2$.

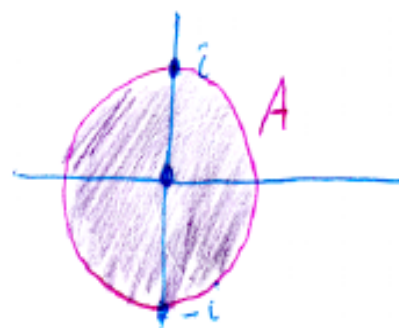
Note. We could have looked at $f(z)$ on the region $2 < |z-i| < \infty$. This would have produced a different Laurent series.

Compute the Laurent series for

$$f(z) = \frac{z+1}{z^3(z^2+1)}$$

on the region $A : 0 < |z| < 1$ centered at $z = 0$.

Solution: This function has isolated singularities at $z = 0, \pm i$. Therefore it is analytic on the region A .



$f(z)$ has singularities at $z = 0, \pm i$.

At $z = 0$ we have

$$f(z) = \frac{1}{z^3} (1+z)(1-z^2+z^4-z^6+\dots).$$

Multiplying this out we get

$$f(z) = \frac{1}{z^3} + \frac{1}{z^2} - \frac{1}{z} - 1 + z + z^2 - z^3 - \dots$$

The following example shows that the Laurent series depends on the region under consideration.

. Find the Laurent series around $z = 0$ for $f(z) = \frac{1}{z(z-1)}$ in each of the following regions:

- (i) the region $A_1 : 0 < |z| < 1$
- (ii) the region $A_2 : 1 < |z| < \infty$.

Solution: For (i)

$$f(z) = -\frac{1}{z} \cdot \frac{1}{1-z} = -\frac{1}{z}(1 + z + z^2 + \dots) = -\frac{1}{z} - 1 - z - z^2 - \dots$$

For (ii): Since the usual geometric series for $1/(1-z)$ does not converge on A_2 we need a different form,

$$f(z) = \frac{1}{z} \cdot \frac{1}{z(1-1/z)} = \frac{1}{z^2} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \dots \right)$$

Since $|1/z| < 1$ on A_2 our use of the geometric series is justified.

One lesson from this example is that the Laurent series depends on the region as well as the formula for the function.

Find the Laurent series of

$$f(z) = \frac{1}{(z-1)(z-3)}$$

in $\{z \in \mathbb{C} \mid 0 < |z-1| < 2\}$.

Proof. We have for $z \neq 1$ and $z \neq 3$

$$\frac{-1}{2(z-1)} + \frac{1}{2(z-3)} = \frac{-(z-3) + (z-1)}{2(z-1)(z-3)} = f(z).$$

For $z \in \mathbb{C}$ with $0 < |z-1| < 2$, we have $\frac{|z-1|}{2} < 1$ and thus

$$\begin{aligned} f(z) &= \frac{-1}{2(z-1)} + \frac{1}{2(z-3)} = \frac{-1}{2(z-1)} + \frac{1}{2((z-1)-2)} \\ &= \frac{-1}{2(z-1)} + \frac{1}{4} \frac{1}{\frac{z-1}{2} - 1} = \frac{-1}{2(z-1)} - \frac{1}{4} \frac{1}{1 - \frac{z-1}{2}} \\ &= \frac{-1}{2(z-1)} - \frac{1}{4} \sum_{n=0}^{\infty} \left(\frac{z-1}{2} \right)^n = \frac{-1}{2(z-1)} - \sum_{n=0}^{\infty} \frac{(z-1)^n}{2^{n+2}}. \end{aligned}$$